



SCIENCE NOTES

CLASS 6



SUMMARIZED SCIENCE REVISION NOTES



SCIENCE REVISION CLASS 6

REPRODUCTIVE SYSTEM

MALE REPRODUCTIVE SYSTEM

Parts of the male reproductive system include

Penis

Testis

Urethra

Sperm duct

Glands

Male parts

Female reproductive system

Parts of the female reproductive include

Ovary

Oviduct

Uterus

Cervix

Vagina

Female parts

Functions

Ovary-produce the ova after every 28 days

Oviduct-where fertilisation takes place

Uterus-where the foetus develop till maturity

Cervix-a ring of muscle that holds the foetus to maturity

Testis-they produce sperms

Urethra-allow passage of sperm and urine in men

Physical changes

In both boys and girls



Increase in height and weight

Hair appear in the armpit and the pubic area

Pimples may appear on the face

In boys only

The voice breaks

They experience wet dreams

The chest broadens

Girls only

The breasts enlarge

Voice become smooth

They experience menstrual flow

Hips broaden

Emotional changes

They mainly affect the feelings, they include;

Feeling shy

Embarrassment of,

Their height

Enlarged breasts

Wet dreams

Menstrual flow

Get moody very fast

HEALTH EDUCATION

IMMUNISATION SCHEDULE

They are diseases that a child is immunised against before they are 5 years.



Most of the diseases are dangerous if they infect a young child.

Immunizable diseases

They are diseases that one can be immunised against;

Typhoid

Yellow fever

Meningitis

Tuberculosis

Measles

Tetanus

Hiv and aids testing

Reasons of getting tested

To overcome fear

To stop the spread of HIV

To plan your future

To start medication early

To decide on marriage partner

To plan for your family

Pre-test counselling; it is the testing one receives in the VCT centre before being tested.

Post-test counselling; it is the testing one receives in the VCT after being tested

Effects of HIV/AIDS to the country

Reduced agricultural production

Poor economy

Death of skilled people

A lot of money is spent on treatment

Congestion in the hospitals

Effects to the family

Sadness in the family



Family income is spent on treatment
Children may become orphans
Children may drop out of school to take care of the sick
parents
The family may feel discriminated.

Effects of hiv to the individual

Poor health
Poor performance
Spending of a lot of money
Feeling stigmatized

PLANTS

PARTS OF A PLANT

Main parts of a plant are;
Roots
Stem
Leaves

FLOWER

flower
The main reproductive part of a plant is a flower.
Parts of a flower include;
Male part
Female part

Functions of different parts;

Male parts

Anthers; produce pollen grain(male cells)
Filament; it supports the anthers

Female parts



Stigma; receives the pollen grains

Style; holds the stigma

Allow the development of pollen tube

Ovary; holds the ovules

Develops in a fruit

Ovule; female cell

Develops into seeds

POLLINATION

it is the transfer of pollen grains from the anthers to the stigma of the same kind

Agents of pollination include;

Wind

Insects

water

Types of pollination

There are two types;

Self pollination-it is the transfer of pollen grains from the anthers to the stigma of the same plant.

Cross pollination-it is the transfer of pollen grains from the anthers to the stigma in different plants of the same type.

Characteristics of insect pollinated flowers

They have nectar

Brightly coloured

Have big petals

Sticky stigma

Small amount of pollen grains

Characteristics of wind pollinated flowers

Small in size



Dull in colour

Powderly pollen grains

Produce large amount of pollen grains

Loosely attached filaments

Germination

It is the change of the seed into a seedling

Conditions necessary for germination

Air (oxygen)

Moisture(water)

Warmth(temperature)

Types of germination

Epigeal germination; the seed comes out of the soil during germination e.g. Beans

Hypogeal germination; the seed remains in the soil during germination e.g. maize

Types of germination

Hypogeal germination

Parts of a seed

Testa

Helum

Micropyle

Cotyledon

Radical

Plumule.

SOLAR SYSTEM

Venus is the brightest and the hottest



Mercury takes the shortest time to go round the sun

Mars is known as the red planet

Jupiter is the largest planet

Saturn has visible rings around it.

Asteroids are found in between mars and Jupiter

Planets ;smallest to largest

Rotation;

it is the process in which the Earth spins on its own axis

The Earth rotates on its own axis

The earth takes 24 hours to make a complete rotation.

The rotation cause day and night

Revolution

It is the process in which the Earth moves around the sun.

The path that is followed by the Earth as it goes around the sun is called orbit

The Earth takes 365 $\frac{1}{4}$ days to make one complete revolution.

Revolution causes seasons

ANIMALS

Animal feeds

They are classified into

Pastures

Fodder

Conserved feeds

Commercial feeds

pasture

They are grasses and legumes that animals feed on directly.

They are classified into;

Pure stand ; only consist of either grass or legumes only

Mixed stand ; consists of both grass and legumes



Grass

Examples of grass include;

Kikuyu grass

Star grass

Giant sataria

Rhodes grass

Legumes

There are 4 main legumes used as pasture, they include;

Clover

Lucerne

Glycine

Desmodium.

Fodder

They are crops that are harvested or cut then given to the animals. examples;

Napier grass

Guatemala grass

Potato vines

Maize stalks

Kales

Sugar beet

Conserved feeds

They are animal feeds that are preserved in a special way to be used in future.

They are divided into two;

Hay

silage



Hay

It is cut and preserved by drying

It is stored in bales

Silage

It is harvested when it is about to flower. It is preserved by fermentation.

It is stored when still green or in the succulent state.

It is stored in silos

The molasses is added to speed up fermentation.

Methods of grazing

Rotational grazing

Zero grazing

herding

Rotational grazing

They include

Tethering

Paddock

Strip grazing

Tethering

The animal is tied to a peg or post using a rope

The rope allows the animal to graze within a restricted area.

It is practised where few animals are kept

Paddock

The land is divided into small areas known as paddocks using a permanent fence

A watering point is usually provided in each paddock.

Strip grazing



The animals are enclosed in a small portion of the pasture using a temporary fence.

An electric fence is usually used.

Zero grazing

It is also known as stall feeding

The animals are confined in a permanent structure (shed)

The shed should have feeding area, watering area, sleeping area and milking area.

Herding

It is a type of grazing where animals are allowed to graze freely on large areas of land.

Water

Waterborne diseases

They are diseases that are spread through contaminated water. They include:

Cholera

Typhoid

Bilharzia.

Cholera

It is caused by bacteria. it can cause death within 24 hours if not treated. It causes death through dehydration.

Signs and symptoms

Violent diarrhoea (rice water).

Vomiting.

Severe abdominal pains

Wrinkled skin due to dehydration.

Sunken eyeballs



Typhoid

It mainly affects the intestines. It is also known as typhoid fever.

Signs and symptoms

Pain in the joints and muscles

High fever

Abdominal pains

Skin rash

Bilharzia

It is caused by bilharzia worms or blood flukes. It is carried by water snails.

The disease mainly affect the bladder and intestines

Bilharzia worms enter the body through the skin.

Signs and symptoms

Blood in urine and stool

Coughing may occur

Abdominal pain

Swimmers itch

Fever

Soil

Soil erosion

Soil erosion is the carrying away of the top soil from one place to another.

Agents of soil erosion

They are things that carry soil from one place to another. They include;

Water

wind



Factors that influence soil erosion

Slope of land

Type of soil

Vegetation cover

Amount of rainfall

Human activities

Types of soil erosion

Splash erosion

Sheet erosion

Rill erosion

Gulley erosion

Splash erosion

It occurs when raindrops fall on bare loose soil.

It can be controlled by;

Planting cover crops

mulching

Sheet erosion

It occurs when water or wind carries away thin uniform layers of the topsoil.

It is not easily noticed

It occurs on gentle slopes.

Best controlled by;

Planting cover crops

Planting trees

Landslides are caused by sheet erosion

.

Rill erosion

It occurs when water flows down a slope and make small shallow channels. The channels are known as **rills**.

it is common on gentle sloping areas.

It can be controlled by;



Terracing
Contour farming
Strip cropping

Gulley erosion

It occurs when water make deep channels , they are known as **gulleys**.

Gulley erosion leads to the formation of V-shaped or U-shaped channels.

It is common on bare hill slopes.

It can be controlled by;

Gabions

Porous dams

Check dams

Food and nutrition

Food preservation

It is the process of storing and handling food properly so as to stop or slow down its spoilage.

Reason for preserving food

To reduce food wastage

To prevent it from being spoilt

For easy transport

To make food available when out of season.

Methods of food preservation

Food preservation is classified into;

Traditional methods

Modern methods

Traditional methods

They include;



Smoking- forms a coat
Drying- reduce moisture
Salting- reduce moisture
Use of honey- prevents oxygen
Use of ash- reduce moisture

Modern methods

Canning-killing germs and preventing oxygen
Refrigeration- low temperature
Freezing- low temperature

Drying is both traditional modern method of preserving food.
It is also the cheapest method of food preservation.

Energy:

Light

How light travels

Light travels in a straight line away from the source.
Light travels to all directions from the source.

Transparent materials

They are materials that allow all light to pass through them
and one can see through them clearly.

Examples:

Clear glass
Clear water
air

Uses of transparent materials

They are used in making:

Car windscreens
Spectacles
Window panes



Lamps

Glass walls

Translucent materials

They are materials that allow only little light to pass through them.

Examples:

Frosted glass

Tracing paper

Oiled or waxed paper

Uses of translucent materials

They are used in making:

Skylights

Toilet and bathroom window panes

Ambulance windows.

Opaque materials

They are materials that do not allow any light to pass through them.

When light hits an opaque materials a shadow is formed.

Examples:

Wood

Stone

metals

Reflection of light

Reflection is the bouncing back of light when Materials that reflect light are called reflectors.

Reflection happens when light hits a smooth shinny surface.

Types of reflection



Regular reflection

Irregular reflection(diffused)

Regular reflection

Irregular (diffused)

Characteristics of the image in a plane mirror

The image is upright

The image is behind the mirror

The image is the same size as the object

The image is laterally inverted.

Refraction of light

It is the process in which light bends or changes direction when it moves from one medium to another. (air to water)

Effects of refraction

Objects appear bent or broken

Objects appear bigger

Swimming pool appear shallower

Making a rainbow

A rainbow is formed by the refraction of light. To be formed *raindrops* and *sunshine* is required.

The process of splitting light into seven different colours is known as ***dispersion***.

A group of seven colours in the rainbow is known as ***spectrum***.

Properties of matter

Composition of air

Air is a mixture of gases

Air mainly consists of :

Gases



Water vapour

Dust particles

Components of air

Components of air

Uses of oxygen

Breathing (Respiration)

Germination

Burning (Combustion)

Rusting.

Uses of carbon dioxide

Photosynthesis

Preserve soft drinks

To make fire extinguishers

Used in baking

Used in making dry ice.

Uses of nitrogen

Used by plants to make proteins

Used to preserve semen

It is taken in through the roots as ***nitrates***. Leguminous plants are able to convert nitrogen to nitrates.

Uses of inert gases

They include Argon, Neon, Helium and Krypton

Used in electric bulbs and light tubes

Used in coloured advertising signboards.

Used in hot air balloons.

Making work easier

Force



Force is a pull, push or lift.

It is measured in Newtons (N)

A moving object is said to be in motion while an object at rest is said to be stationary.

Force is measured by the use of a **spring balance**.

Examples of force

Force of gravity (weight)

Frictional force

Magnetic force

Inertial force

Effects of force

Makes an object to start moving

Stops a moving object

Change direction of a moving object

Speeds up a moving object

Change the shape of an object.